

Project Statement

The **Personal Expense Tracker** is a Python-based application developed to provide a simple, efficient, and organized solution for managing personal daily expenses. In day-to-day life, keeping track of small and large expenses manually can be difficult and time-consuming. This project aims to provide a digital method for recording and managing financial transactions in a structured manner.

The application allows users to add expense records by entering important details such as the amount spent, expense category, description, and date. Each expense is assigned a unique identification number, making individual records easy to manage. Users can view all their recorded expenses, calculate their total spending, and analyze their expenditure based on different categories such as food, travel, shopping, bills, education, entertainment, and other expenses.

The system also provides search functionality that enables users to find expenses based on their category. Unwanted or incorrect records can be removed using the delete option. To ensure that the recorded information is not lost when the application is closed, the project uses a **JSON file** for persistent data storage. This makes the application lightweight and easy to use without requiring a separate database system.

The project is developed using **Python 3** and demonstrates the practical implementation of important programming concepts such as functions, conditional statements, loops, lists, dictionaries, user input, exception handling, file handling, JSON data management, and date-time operations. The project can be further enhanced in the future by adding graphical interfaces, budget tracking, monthly reports, charts, database connectivity, and PDF or Excel report generation.

Overall, the **Personal Expense Tracker** provides a practical and user-friendly approach to maintaining personal expense records while demonstrating how fundamental Python programming concepts can be combined to develop a useful real-world application.